

Electricity Consumption Forecasting Project

1. Introduction

Electricity consumption forecasting is crucial for energy management, grid stability, and cost efficiency. This project aims to predict electricity demand based on historical consumption data and weather conditions. By leveraging time series analysis and machine learning models, we aim to enhance energy planning and reduce operational costs.

2. Dataset Description

The dataset, sourced from Kaggle, consists of electricity usage data recorded at **5-minute intervals** along with weather variables such as temperature, humidity, wind speed, and pressure. The key columns include:

- **Timestamp:** Date and time of the record.
- **Power Demand (MW):** The electricity consumption at a given timestamp.
- **Temperature (°C):** Ambient temperature at the time of recording.
- **Humidity (%):** Percentage of moisture in the air.
- **Wind Speed (m/s):** Speed of the wind at a given time.
- **Pressure (hPa):** Atmospheric pressure.

3. Exploratory Data Analysis (EDA)

3.1 Correlation Analysis

The correlation heatmap shows:

- **Positive correlation** between power demand and temperature (**0.40**).
- **Negative correlation** between power demand and pressure (**-0.39**), indicating higher electricity usage in low-pressure conditions.
- Humidity has minimal direct correlation with power demand.

3.2 Time Series Visualization

- **Daily Power Demand Trend (2021-2024)**: Displays seasonality and fluctuations.
- **7-day Moving Average**: Highlights smoothed trends and seasonal variations.

4. Data Preprocessing

Steps taken for preprocessing:

- **Handling Missing Values**: Interpolation was used to fill missing values in weather data
- Resampling data 5 min interval to day to day
- Fixing spiky plot
- Rename columns

5. Forecasting Methodology

5.1 SARIMA Model

After experimenting with multiple models, **SARIMA (Seasonal AutoRegressive Integrated Moving Average)** was selected due to its ability to capture both trend and seasonality effectively.

Why SARIMA?

- The dataset exhibited **clear seasonality** due to periodic variations in power consumption.
- SARIMA integrates **differencing** to handle trends, **autoregressive (AR) components** to model past values, and **moving average (MA) components** to smooth fluctuations.

SARIMA Model Configuration

- **(p, d, q) Parameters:** Determined using ACF/PACF plots and grid search.
- **(P, D, Q, s) Seasonal Parameters:** Optimized based on seasonal patterns.
- **Model Training:** Data was split into training (80%) and testing (20%).

6. Model Evaluation

The SARIMA model's performance was evaluated using:

- **Mean Absolute Error (MAE): 287,249.90**
- **Mean Squared Error (MSE): 139,283,540,200.87**
- **Root Mean Square Error (RMSE): 373,207.10**
- **Mean Absolute Percentage Error (MAPE): 38.12%**

The relatively high error rates suggest that while SARIMA captures seasonal trends, further improvements (e.g., hybrid models) could enhance accuracy.

7. Results & Interpretations

- Power demand fluctuates seasonally, peaking during extreme temperatures.
- Weekday vs. weekend demand variation was observed, with higher usage on weekdays.
- Weather conditions significantly influence electricity consumption.

8. Conclusion & Business Impact

Accurate forecasting enables:

- **Better Energy Planning:** Power grid stability and efficient energy distribution.
- **Cost Optimization:** Reducing wastage and optimizing power generation schedules.
- **Demand-Supply Balancing:** Helping utility companies prevent shortages and overloads.

9. Future Scope & Improvements

- **Anomaly Detection:** Identifying power outages or unusual consumption spikes.
 - **Real-time Forecasting:** Implementing models for live data streaming.
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